

Introduction to Artificial Intelligence and Text Processing

A Sample Document for Natural Language Processing Labs

1. Overview

Artificial Intelligence (AI) is the branch of computer science that builds machines capable of performing tasks that normally require human intelligence. These tasks include learning, reasoning, problem solving, perception, and understanding language. Today, AI is used in healthcare, finance, education, transportation, and entertainment.

Natural Language Processing (NLP) is a subfield of AI that focuses on the interaction between computers and human language. NLP systems can read, interpret, translate, and even generate text. Common applications include chatbots, search engines, spam filters, machine translation, and sentiment analysis!

Before any NLP model can work with text, the text must be cleaned and prepared. This is called PRE-PROCESSING. Typical steps are: lowercasing, removing punctuation, removing stop words, tokenization, stemming, and lemmatization. Have you ever wondered why the words running, runs, and ran are treated as the same idea? That is exactly what stemming and lemmatization solve.

2. Key Concepts

- Tokenization: splitting text into words or sentences.
- Stop words: very common words such as the, is, at, which, and on.
- Stemming: reducing words to a root form (studies, studying, studied -> studi).
- Lemmatization: reducing words to a dictionary form (better -> good).
- Bag of Words: representing text as counts of each word.
- TF-IDF: weighting words by how important they are in a document.

3. Contact and Reference Information

For questions about this course, contact the instructor at instructor@university.edu or the teaching assistant at ta.support@university.edu. Phone: +92-51-1234567. Visit the course website at <https://www.example.com/nlp-course> or read the documentation at <https://docs.python.org/3/>.

Important dates: The lab was published on 15/09/2026, the first assignment is due on 2026-10-05, and the final exam takes place on October 20, 2026. Course code: CS-4021. Room number: B-204. Total budget for the lab: \$1,250.75.

Regular expressions can find patterns such as emails (name@site.com), dates (2026-10-05), phone numbers (+92-51-1234567), hashtags like #NLP #AI #Python, and mentions like @student123.

4. Sample Paragraph for Word Frequency

Data is the new oil. Data drives decisions, and good data drives good decisions. Machine learning models learn from data, and the more data a model sees, the better it can learn. However, bad data leads to bad models. Clean data, clean code, and clean thinking are the keys to success in data science.

5. Sentiment Examples

I absolutely love this course! The lectures are clear and the labs are fun. The instructor explains everything wonderfully.

This assignment was terrible. The instructions were confusing and the deadline was too short. I am very disappointed.

The library is open from nine to five. Students can borrow up to five books at a time.

6. Table: Student Exam Results

Student ID	Name	Subject	Marks	Grade
S001	Ayesha Khan	Mathematics	92	A
S002	Ali Raza	Physics	85	B
S003	Sara Ahmed	Chemistry	78	C
S004	Usman Tariq	Computer Science	96	A
S005	Fatima Noor	English	88	B
S006	Hamza Malik	Statistics	71	C

7. Table: Quarterly Sales (in thousands)

Quarter	Product A	Product B	Total
Q1	120	80	200
Q2	150	95	245
Q3	170	110	280
Q4	210	130	340

8. Conclusion

In this document we introduced AI, NLP, and common text pre-processing steps. We also included emails, URLs, dates, tables, and sample sentences so that you can practice text extraction, cleaning, tokenization, word counting, regular expressions, table extraction, and sentiment analysis. End of document.